

Q1: 17 March (Shift 2) - Single Correct

During which of the following processes, does entropy decrease?

(A) Freezing of water to ice at 0°C (B) Freezing of water to ice at -10°C (C) $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$ (D) Adsorption of $\text{CO}(\text{g})$ and lead surface(E) Dissolution of NaCl in water

(1) (A), (B), (C) and (D) only

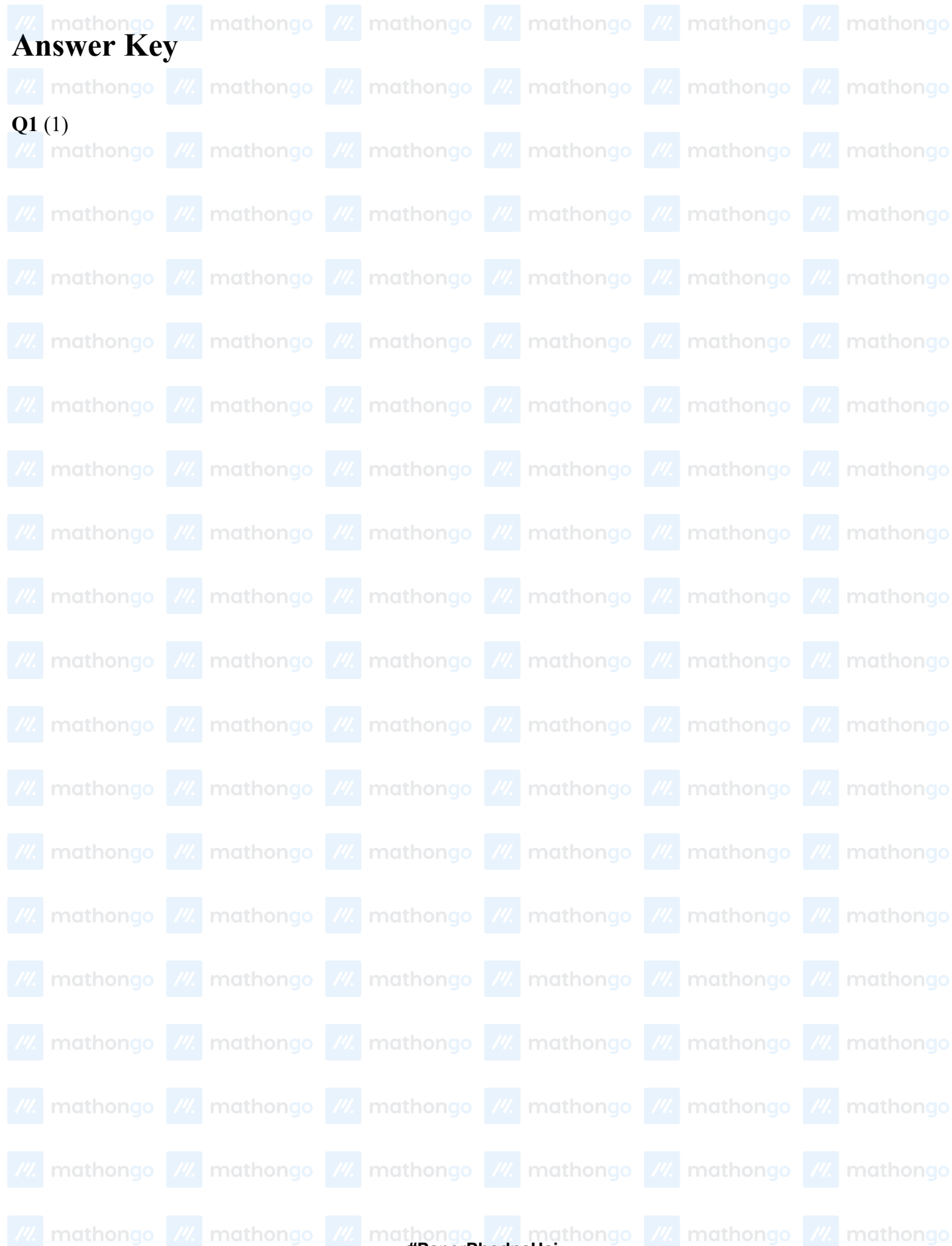
(2) (B) and (C) only

(3) (A) and (E) only

(4) (A), (C) and (E) only

Answer Key

Q1 (1)



Q1 (I)

(A) Water $\xrightarrow{0^{\circ}\text{C}}$ ice; $\Delta S = -\text{ve}$ (B) Water $\xrightarrow{-10^{\circ}\text{C}}$ ice; $\Delta S = -\text{ve}$ (C) $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g}); \Delta S = -\text{ve}$ (D) Adsorption; $\Delta S = -\text{ve}$ (E) $\text{NaCl}(\text{s}) \rightarrow \text{Na}^+(\text{aq}) + \text{Cl}^-(\text{aq}); \Delta S = +\text{ve}$